Running Out of Matter with Michelle Thaller

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis The episode explores the precarious intersection of cutting-edge astrophysical discovery and the systemic socio-economic threats facing scientific research in the United States. While human curiosity continues to drive breakthroughs in understanding complex phenomena—such as gravitational waves, neutron star dynamics, and the expansion of the universe—the stability of foundational institutions like NASA is being undermined by budget volatility and "brain drain." The discussion posits that the loss of scientific talent to international competitors (Europe, China, Canada) due to domestic funding instability represents a significant threat to American technological and economic leadership.

2. Key Arguments & Data * **NASA Budgetary Disparity:** The science-specific portion of NASA's budget is approximately $6 billion out of an annual total of roughly $30 billion. This amount represents less than 0.4% of the total U.S. tax dollar distribution, a figure so small it is visually imperceptible on a single one-dollar bill. * **The "Brain Drain" and Economic Impact:** Budgetary uncertainty and the termination of "probationary" employees (those with less than three years of tenure) are driving top-tier American scientists and engineers to overseas institutions in Germany, France, and China. This is economically detrimental because science research has been shown to yield 2–3 times the economic advancement of the initial capital invested. * **Gravitational Wave Precision:** The LIGO (Laser Interferometric Gravitational Wave Observatory) project demonstrates extreme engineering capability by detecting changes in spacetime distance equivalent to measuring the width of a human hair across a distance of several light-years (e.g., 24 trillion miles). * **Neutron Star Extremes:** Neutron stars pack roughly twice the mass of the Sun into a diameter of only about 20 miles. Their intense magnetic fields facilitate "synchrotron radiation," an event where high-energy particles are accelerated, allowing high-energy cosmic events to be detected as lower-frequency radio waves. * **Cosmological Entropy:** The universe is moving toward a state of "heat death" (or "cold death") characterized by increasing entropy. As the universe expands, photons undergo "redshifting," where their wavelengths lengthen and energy levels

drop, eventually leading to a universe of low-energy, undetectable radiation. * **The Physics of Light in Media:** When light passes through a medium (like water or glass), its effective speed changes due to the interaction between electromagnetic fields of the photon and the atoms/molecules in the medium; the light is not necessarily "slowing down" but rather being temporarily delayed by interacting with surrounding electric fields.

3. Actionable Takeaways * For Aspiring Scientists: Prioritize "grit," ambition, and persistence over a perceived lack of innate talent. View complex subjects like mathematics and physics as a language; mastery is achieved through long-term commitment and fluency rather than immediate, effortless understanding. Embrace the "nerdy" identity as a modern strength. * **For Educators/Science Communicators:** Use metaphors (e.g., learning a language, pool balls in a medium) to demystify complex physics. Focus on fostering curiosity to ensure that students do not become discouraged by the difficulty of STEM subjects. * **For Citizens and Policy Advocates:** Recognize that scientific funding is a high-yield economic investment. Support for agencies like NASA and the NIH (National Institutes of Health) is essential for maintaining national competitiveness and addressing critical issues like climate change, planetary defense, and medical breakthroughs.

Full Transcript

Chuck, we finally got Michelle Fowler on StarTalk, a friend, a colleague, and science educator. We love these folks. Oh, yeah. She's fun. She's got the energy that we all want and seek of someone who's enthusiastic about the moving frontier of the universe. The energy of a neutron star in the compact form of an astrophysicist. Ooh. That came out of the show. I got that from the show we just did. That's the highest compliment I've ever heard anyone get who's an astrophysicist coming right up on Star Talk. Welcome to Star Talk. Your place in the universe where science and pop culture collide. Star Talk begins right now. This is Star Talk. You're the grass Tyson, your personal astrophysicist. Got with me, Chuck. Nice. How you doing, man? What's up, Neil?

All right, we're doing Cosmic Queries today. Yes, we are. With an old friend and colleague, Michelle Thaller. Michelle, welcome to Star Talk. It is wonderful to be here. Great, great to be talking to you guys. Yeah, so you're not only a fellow astrophysicist, you're a fellow science communicator. And we love that species of scientists out there. You are recently... of NASA's Goddard Space Flight Center. But the last time you weren't recently of it, you were of it. So how recently have you no longer been part of them? Yeah, I retired in October, I think it was October 2024. So yeah, yeah. So I've been retired for a couple of years. Been doing speaking. I've been doing a lot of lecturing with Smithsonian Journeys. That's been a lot of fun. I know that, yes. Yeah, just having fun being a freelance astrophysicist. There you go. Nice. Whoa, astrophysicist at large. We got a personal astrophysicist and a freelance astrophysicist. We can talk about whatever you want. I'm not working for anybody.

for anybody. Very, I like that. But let me just finish up with your CV here. You were research scientist with the Spitzer Space Telescope, one of the, one of the big, what do they call them? The, the grand. Great observatories, yes. Great observatories. And that one was a band of the spectrum that Hubble did not cover. So that was infrared, right? The Spitzer Telescope. Could you straighten us out here? Locally, we have the Goddard Institute for Space Studies who lost their lease. with Columbia as it was removed by

the White House, but that has more Goddard in it. And then we have the Goddard Space Flight Center in Maryland. They're both NASA. Could you just help us distinguish the two? Well, yes. So the Goddard Institute for Space Studies has for a long, long time been the center for NASA, basically Earth science, climate science. Yes.

And so, you know, that was the thing. So they had a very close link to Columbia University. Columbia University leased this building to them. Famously, it's the building that, if you remember, Seinfeld, you know, the sitcom, Seinfeld, it's the cafe. That's that building. Yeah, that's right. The diner's in that building. That's a massive building. It's the greatest name for any eatery ever. It's a restaurant. Yeah. Yeah. So, I mean, absolute heroes of mine work there. I mean, these are the people that are doing, you know, they're in the middle of Antarctica, ground-truthing measurements from satellites, you know, they're making sure we understand what's going on with, you know, the atmosphere and the oceans and all of that. You could say it, Michelle. You could say, they're doing God's work. You could say that. God's work.

God is doing God's work. Absolutely. No, two heroes of mine. And yeah, it's been rough. It's been a rough couple of years for them. Yeah. In fact, at the museum, we have a few refugees that have taken pitched tent within our facility because we're just down the street a couple of miles from Yes, Goddard Institute for Space Studies, but they're homeless at this point. They're still all paid as scientists, but they're homeless. But it has the name Goddard in it, yet so did the Space Flight Center. So now we go to Maryland, then what happens there? Well, yeah. So Maryland, right. So this is this giant base. At the moment, it's still NASA's largest base in terms of people that go there. I think they're down a little bit. But when I was there, there were about 10,000 people that worked there every day.

And that's pretty incredible. I mean, one statistic they had is that if Goddard were to leave NASA, then Goddard would become the world's second largest space program. I mean, Goddard itself is bigger than the other space programs around the world. So Goddard builds satellites or telescopes that get launched. Well, yeah, well, so, I mean, the Hubble Space Telescope was built largely at Goddard, the observatory part of the James Webb Space

Telescope. We build the GOES satellites, the weather satellites. I mean, yeah, I mean, there's four huge science divisions. You've got Earth Science, which is the biggest one. And then Heliophysics study the sun, the Space Weather Action Center, all the stuff the sun's throwing at us. And then you've got all the planetary stuff and the astrophysics stuff. It's the world's largest science space that we know of.

Yeah, for now. Please keep it that way. Yes. We'll get there, Chuck. Yeah. Michelle, tell me what are the consequences of the budget cuts we've read about? to NASA overall and to Goddard specifically, where you once worked. It's been a couple years of, I mean, I really feel for all of these people, all of my former friends, and so many of them have left. I mean, I've seen some of our absolute best scientists, and not only that, our best engineers, one of my best friends who's just an incredible kick-ass systems engineer. She got scooped up by Switzerland. And i mean the scientists are going overseas with nasa and god in general it's been this real seesaw is there gonna be a huge cut you know or well then congress kind of restores the funding so maybe not oh no no we're gonna do the huge cut not so i mean it's this time of terrible uncertainty and i mean aside from your programs being on the cancelling block and losing scientists.

The other big thing I've been seeing that's been heartbreaking is the best young people, the best scientists. So NASA was required to get rid of all the people that were hired in the last three years, all the people that were still on probation. This may be the person you wanted to get, the leader. Right out of graduate school, they're going to be the leader for the future. If they were there for less than three years, you can still legally fire them. And so those people are gone. And you have a lot of the best scientists now going, well, why should I work for NASA? Why should I work for any kind of government agency? Because they can just up and cancel my job at a moment's notice. And just to be clear, when you use the word probation, you don't mean they did something bad and they're under observation. This is just the starter period of time over which you could be removed for no cause at all just because you're not vested in the in the government system yet. Is that a fair way to characterize that? That's right. I mean, everybody sort of has a trial period when you're hired as a government scientist, equivalent to sort of like getting tenure at a university. You're there sort of

for a while where it's legally easy to say, oh, well, that wasn't the right fit. Maybe you should go somewhere else. But then when they make you permanent, that's harder to do. And so yeah, by probationary, that doesn't mean that anything, nothing was wrong. These are the best people in the world. If you do something wrong, they put you on double secret probation.

So Michelle, this sounds like when we, I think I might be a little older than you, but same sort of generation where we're in graduate school and we as a nation are the beneficiaries of the smartest people in other countries coming to the United States for opportunity to express their talents and their brilliance. and said, bad for their home country, good for America, right? And now I have colleagues, as do you, who are getting the phone call from Europe. We got a lab for you in Germany. We got a lab for you in Paris. And here's money, and it's funded, and it's stable. Come on down. And so now the world is cherry picking us. They weren't sending us their best, their brightest. They weren't.

Yeah, but now we're sending our best and brightest out. Yeah, yeah. So, so where is this land do you think? Are we going to recover from this? Well, I mean, one of the physics journals, I remember had a full page ad from the country of Denmark saying, you know, come to a place where facts still matter, right? Oh, yeah. Yeah. No, I mean, other countries are on it. And Yeah, Europe is a beneficiary, Canada, and China. I mean, China as well. I have a lot of colleagues that got offers that will bring your whole lab over, all your graduate students, like you said, funding. It's so unnecessary. I mean, the thing that has been so tragic about this.

is that science research in the United States is a low budget item. Oh my God. We do not spend a lot of money on this already. I mean, the thing that I was so proud of, I mean, I worked for sort of the science part of NASA, doing all the earth science, the Mars rovers, the web telescope. About 110 active missions at any given time. Each of those missions employs thousands of people. In some cases, tens of thousands of people across the country, across the world. they do it all for under six billion dollars a year. I mean, that's a nothing burger when it comes to a budget. I mean, we get so much innovation. You know, you can demonstrate, we've had economists do studies that you get two, three times the economic advancement of

money put into science research. Just to be clear, that's the six billion out of the 30 billion that is NASA's annual budget. And you're referring to just the science, which is not putting people in orbit. That's not the space station. That is not launching astronauts. That is science. Putting people in orbit, of course, has to be more expensive because you need to be a lot more careful when there are human lives on stake. They want to come back usually. They want to come back. The rovers don't need to come back. People, you got to bring them back. What you can do for the cheap? I mean, some of the missions that blow me away.

you know, I mean, basic physics had a neutron stars work nicer. That's actually on the space station, but it's a small thing, just a couple hundred million dollars. I mean, these little budget items, you know, James Webb, of course, is a bigger one, but that was over like 20 years and employed again, tens of thousands of people. You know, these are these are tremendously great value for the money. Did you say it was 30 billion? That's all we spend 30 billion in the total. Yeah, it's around there, plus or minus for NASA in a year. Well, 30 billion is less than four-tenths of 1% of your tax dollar. And I do the experiment, you take a full dollar and just cut into it, just a single dollar bill, cut into it four-tenths of 1% of its width. And it doesn't even reach the paint, the ink of the bill. Okay, you can remove that and the bill is still there. A, B, what I wanted is to create a new budgeting system where agencies get the money people think they're getting. The visibility of NASA is so huge compared to their actual budget. No one would guess off the top that they're getting four tenths of one percent. They would say five percent. I've heard some people say ten percent. And I say that's the budget we should give them. Give them their visibility budget.

Yeah. Yeah. And by the way, everybody loves NASA still. It's one of the few government agencies where people believe in the institution of the agency and they also trust it. And it's respected around the world. It's respected around the world. You know, that and what was what was formerly the NIH, which is now the SAD, as in just sad. That doesn't even stand for anything. It's just sad. I mean, NIH is a whole other story. I mean, honestly, that's even more worth saving than NASA. I mean, as much as I'm a huge NASA and space fan, I mean, do not mess with the NIH. Good Lord. I mean, we need to keep that research going, the cancer research, the

Alzheimer's research. I mean, again, the Center for Disease Control, things like that.

Oh my God, do we need those? And remember, for 30 billion, you're still talking about the whole budget of NASA, and the human space program is the big chunk of that. All the earth science, all the climate science, all the weather satellites, all the monitoring the sun, all the exploring the universe. And all the space telescopes are in the sun. All the space telescopes. Yeah. I mean, I don't know if this is true, but I remember one economist saying that they'd shown that the Pentagon, the military across all over the world had spent more money on air conditioning in a year than that budget. Oh my God. So I'm not absolutely sure that's true. But, you know, it's on that level. I mean, we're not even a fighter jet, basically. Looks like America has stepped in a pile of doge crap. Yeah, doge do do.

This is Ken the nerd neck Zabara from Michigan and I support star talk on patreon This is star talk radio with Neil deGrasse Tyson So let's pivot now to our question so Michelle we've solicited questions from our fan base well from our patreon supporters and they know who you are they know your expertise and These are targeted for you and I to the extent that we overlap in expertise I'm happy to put in my two cents, but we're really here to see

what you've got to say in response to these questions. Chuck has them all. I haven't seen any of them. And so Chuck, what do you have for us? All right. So, inquiring minds want to know. This is Ross Graves, who says, hello, I admire and I respect you all, Dr. Tyson, Dr. Tholler, Lord Nice. This is Ross from California. When a photon is absorbed and re-emitted or reflected, is the journey through space-time physically continuous, or does that interaction create an entirely new quantum history for the outgoing light? in the sense that experience is no time of its own. If reflected, did it hit a surface and restart its experience as a reflection? Dude, that's a matrix like crap. He just can't. That is an amazing question. I love it. I love it. Did it take on a new identity as a reflected photon? That's what that comes down to. So, Michelle, what do you have to say about that?

Okay, all right, all right, so I'm gonna say, I'm gonna start out saying this is something that is a question of mine. So I mean, well done, because I mean, this is actually something that I wish I understood better. I only recently

came to a better understanding because of some great podcasts actually. It was a podcast Veritasium that Derek Muller does. that finally I could understand why the speed of light is different speeds. When you go through the vacuum of space, it's the fastest anything can go, the speed of light. When it passes through air or water or glass, you see that classic bend, right? When you look through water, you see like bend, and that's because the speed changes. And it was only his podcast that finally got me to understand, light is an electromagnetic field.

And when it's in the presence of other electromagnetic fields, when you're not in a vacuum, there's other atoms and molecules around, the electromagnetic field basically becomes a combination of those two fields. And he did this great podcast about that. The question about whether a photon is reflected becomes a different photon. It's fascinating. So let's break it up. Let's go back to the beginning. I mean, this is one of the craziest. And it's true. The idea that when you're traveling at the speed of light, I mean, photons obviously don't have any consciousness that I'm not able to observe this. But if a photon has no idea that space and time exist, when you're traveling at the speed of light, time goes to a zero. You don't experience the passage of time. And when you think about you're frozen in a moment of time, you can go any distance in the universe instantaneously to your perspective. So a photon doesn't really experience the universe as ever having expanded. We're still a point, a thing.

you know, almost a singularity to a photon. It does not experience space and time. What the hell does that mean? I mean, how do you wrap your head around that? I mean, the light that's bouncing off my lamp from my face right now doesn't agree with me that space and time exist. Now, help me unpack this, Neil, because I don't have a great answer. I would like a better answer. Let me restate what you said, but I don't have a deeper understanding just for having restated it. It's the photon has to know that the universe expands because it gets redshifted, right? So the photon you detect has a different energy than the photon that was emitted. So even though the photon does not experience space time, we measure a different photon, a different energy for that photon.

some of that energy got shared with the expansion of the universe. So that's something I can't claim to fully understand. How it exists doesn't exist in time at all, yet it changed its wavelength before it arrived on location with you. Now let me ask you, I didn't see that episode with Veritasium, but it was also a recent understanding of mine as well, that when photon moves through a medium, It is moving at the speed of light between atoms of the transparent medium, but it interacts with each atom and continues forward, and that interaction creates a time delay. So the photon is never actually ever going slower than the speed of light. It just got waylaid.

on route through the transparent medium. Now, is that consistent with how, what you got out of the Veritasium? So wait, wait, Neil, let me just, because I'm, wow, Jesus. So I'm trying to, I'm trying to. Take a cleansing breath, Chuck. Yes. Take a cleansing breath. Okay. I'm trying to get this as a metaphor because that's how I see and understand this. So in the vacuum of space, that would be like, taking a pool ball and shooting it straight across the table into the pocket. And it's going at that speed. But then when you put it through a medium, that would be like having the same speed, but banking it off of three banks and then going into the pocket. So your speed doesn't change, but the fact that you had to bank it three times is what slowed you down. However, however, what you're describing does happen to light, but that's when light scatters.

Okay, so that's a different phenomenon. It keeps in the same direction at all times. I gotcha gotcha gotcha. That's the difference between a transparent medium, a translucent medium, and just something that completely scatters the light. So so so Michelle, are we an agreement that we're talking about a new photon as it goes through a newly shaped photon after it interacted with these other fields? Is that a fair? The way the podcast did it, and this was the first time I ever sort of thought, oh, okay, I get this. We're sort of talking about photons as particles here, like little billiard balls and stuff, but of course, remember, famously, they're electromagnetic waves, right? I mean, they're not just little balls. And so what the podcast did is it showed that there's electromagnetic waves, there's a photon going in, and there's other atoms around.

And basically the electromagnetic wave starts those other atoms kind of vibrating along with it. But there's a delay, there's a lag to kind of get those other fields, those other electric fields vibrating as the photon goes by. And the resulting speed is actually a combination. It's actually a way of resolving those two different vibrational modes. So the photon is like to the other atoms like, bro, you're bringing me down. You're a drag on me man. I can't I got it come on. I'm serious like get off me and then let me get the hell out of this medium Yeah, because once he exits the medium is back at the speed of light The speed of the wave that we measure is a combination of the speed of the photon in a vacuum But then also this the speed of the variations of the of the of the basically the the oscillations of everything around it those two combine into the measures we see. So the question about, I mean, it's the same thing, Neil. Absolutely. When you talk about how a photon changes, even as it goes through a vacuum, it loses energy. It's a very similar question that if a photon is reflected, is it a different photon? I mean, it could certainly be a different energy, right? I mean, when a photon is reflected, you usually lose some energy in the reflection. So, I mean, there's all sorts of things photons absolutely can change.

Just to be clear, Chuck, if the photon hits a surface and reflects, some of its energy is imparted as momentum to that surface, and that's why solar sales work at all. Yeah, yeah. Yeah. So yeah, there's an energy change there. So I'm voting for a completely different photon here. What do you think, Michelle? Yeah, I like I said, I Ross graves is a name. That's familiar to me. So yes, I think they thank you for this fabulous question and I have to say I do not have an extremely clear answer I'm not ashamed of that. This is what I've wondered myself. Okay, so let's let's vote I'm voting that the photon that came out is a new quantum state that is carrying the mission forward after it's a delay from the medium through which it passed or the medium off of which it reflected. I'm going with that. I wish I had the same properties and there was a reflection chuck and real chuck and I could tell the All the people I owe money, you need to go talk to Reflection Chuck. That's not me anymore. That's not me anymore, buddy. No, that's a great question. OK, what else do you have, Chuck? All right, here we go. This is Jal Costa. And Jal Costa says, good tidings, Dr. Tyson, Dr. Fowler, Lord Nice, deep breath now, Chuck. Jal Costa here. Greetings from Portugal. All right. Portugal.

Yes, he says, if matter isn't being created, only transformed, will there come a day when the universe runs out of matter to create new astral bodies? If so, how will we be able to notice such a situation developing? Will the creation start to become rarer, smaller, or deficient as in less frequent, less energetic, or less stable? Thank you and please keep making our necks hurt. Oh, I like it. Look up, making that hurt. Oh, oh, no. I see. I see what he did there. Okay. You got him looking up so much you gave him a neck ache, Neil. Okay, you can lay down and look up and then you just looking straight ahead. The geometry. Yeah. Yeah.

Well it sounds to me like somebody is just describing what's called the heat death of the universe and the answer is yes. The answer is that the universe is slowing down in the sense that It's not so much that matter can't be destroyed. It can. It can be converted into energy. Matter and energy are two sides of the same coin. That famous equation, hopefully the only one you ever have to deal with in life, unless you want to be a scientist, is energy equals mass times the speed of light squared. Good old Albert Einstein. E equals mc squared. That's most people's first equation they learn in elementary school. You can destroy matter, you can make it into energy, and you can make energy into matter. That's what we do in particle accelerators. We get things going with such high energy when things collide, there's so much energy around it actually creates new matter. That's how you find new types of particles. So energy and matter are equivalent.

And so there is, but absolutely what's happening is that in the early universe there were things that were much higher temperature, much higher energy. Stars, for example, are an example of something that's a very low entropy. Stars create very high energy radiation. And, you know, over time, yes, I mean... Every time you basically generate heat, sunlight comes down, it's a high energy photon, hits the earth, warms things up. What's released, the heat is a lower energy type of light. You've lost energy, you've made from a high energy particle of light, now it's a lower energy type of light. And so eventually the universe, all of the photons that were high energy are gonna hit things, get absorbed by things, make things warm, get re-emitted, and they're gonna lose energy over time.

And so, yes, the universe is running down. And eventually we're going to get to a point where there's nothing left but very low energy photons. Photons that are so low energy, they're probably not even detectable. And then it's sort of anybody's guess as to, you know, does time and space exist anymore when there's really, everything is just the same temperature, a very, very low temperature is the only thing that the universe has. So it's a misnomer to quality heat death if it's actually a cold death. It gets cooler and cooler. It's the heat that's dying. It's the heat that's just like, go on without me. Just go on without me. Plus the expanding universe also drops in dropping the temperature with it, right? Absolutely. Got a photon going through expanding space and think about, you know, draw a squiggly line on a piece of elastic and pull the elastic and then that wavelength drops down. You lose energy. You know, I mean, it's really kind of as simple as that. Wait, just to be precise, the wavelength gets longer. The wavelength gets longer. That's right. Yes. And that's a lower energy wavelength. Lower energy. So it drops down. The wavelength gets longer. The frequency drops. Right. And if I may, what Michelle just described, is exactly how the warming of the planet happens high energy comes in pierces our atmosphere becomes lower energy and a longer wavelength which is trapped by greenhouse gases and that's how we heat up because as everybody has learned from meal the atmosphere isn't hot.

the ground gets hot and radiates heat back into the atmosphere. And that is the cycle that creates a warming of the planet. Just so that you guys know that it's real. Okay, it's real. And send 5 to the Chuck Stennett campaign. By the way, just to be precise, there is a layer of the atmosphere that does absorb light directly from the sun. And that's the thermosphere where the ultraviolet gets absorbed. and by the ozone. And so that there is a layer that is hotter, but we're not experiencing that right down here. Right. Oh, yeah. That's cool. Yeah. The hottest part of the troposphere is right above Earth's surface. Right above Earth's surface. Okay. Nice. Yeah. Right. And what's the part? See now, now this is where the explainers are all swirling in my head. What is the part where the excitation of the molecules is such that It's, we say it's cold, but it's actually hotter. Chuck, could it be that there's a layer, upper layers of the atmosphere where the particles themselves are high energy, but there isn't many of them. Right. So that you don't feel that

temperature that you otherwise would. That's what I'm saying. So the concept of temperature kind of loses practical meaning there.

Right, okay. I said it like a what you think about yeah, that's exactly what I was thinking about but I said it like a dumb two-year-old If I if I flew you through the Sun's corona Which is five million degrees? Would you feel that given how low? Density the particles are that have such high temperature We are really really lucky the answer is no because I mean the corona is one thing but but also let's remember I mean we actually our whole solar system is in this hot bubble of gas in the Milky Way galaxy. Maybe there was a star that exploded here billions of years ago, but the temperature of the gas we're going through is in the millions of degrees in our galaxy. The thing is, the gas between the stars is so thin. It's like, okay, there goes a proton, zoop, there goes an electron, zoop. I mean, there's so few particles that it doesn't really impart any heat to you.

I mean, yes, the average speed of these high-energy particles is equivalent to being millions of degree gas. And just like Neil said, the sun's corona is this extended atmosphere of very hot gas. We have a spacecraft, NASA has a spacecraft called the Parker Solar Probe, that is orbiting well within this solar corona right now. You better believe it would not be a live returning data if it really felt like five million degrees. Yeah. So the analogy we used to use is when you think about everybody who's done some baking, you get your oven up to 500 degrees. That's some serious baking if you're baking the 500 degrees. It's pizza. We're going for pizza ovens here. You're a pizza shop on the corner of Lenox Avenue 125th Street. If you have a 500 degree oven, OK. So if you open the oven and you just stick your hand in, your hand is not actually feeling like it's 500 degrees all of a sudden. If you were to touch like a metal part of the oven, if you were to touch the rack, then you'd burn your hand. Because I mean, all of a sudden now you have a denser object. You've got the metal.

A little bit of a thing about the Parker Solar Probe. It's another incredible person, Eugene Parker, and he was the person that discovered this wind of pine energy particles from the Sun, the solar wind. The solar wind. Yeah a fabulous launch i'm glad somebody told me that a delta heavy looks like it explodes on the pad before the rocket comes up i was at that launch is

amazing the whole idea the surface of the sun is only about you know say roughly like ten thousand degrees Fahrenheit but as soon as you get off the surface the gas becomes millions of degrees hot and i mean this was a big. mystery. And how does the sun do that? I mean, normally, if you've got a hot campfire, the farther away you walk, the cooler it seems. Why does the temperature go up? And the Parker Solar Probe has really helped answer that. There's all of these complicated accelerations due to magnetic fields around the sun. The particles pick up speed. There's shocks in there. So I mean, we're figuring out how the sun works. That's not something we know until we go there near the sun, fly around in this corona.

and see what's going on. And look at that. You're getting all of that for a lousy six billion dollars. What the hell is wrong with your people? OK, Chuck. That's his daily I got to blow a gasket once a day. Reaction. Worth it. Worth blowing a gasket over. If memory serves, the Parker Solar Probe set a speed record. Oh, yeah. It goes over 400,000 miles an hour. Oh, yeah. Wow. Yeah. I keep having to look that up because I don't believe it. And then I look it up again. Because it gets pulled in by the sun's gravity. And there it is, 400,000 miles an hour. Wow. Amazing. All right, let's move on. I have a few more questions. Let's try to fit in a bunch more. So let's tighten up the answers and see how many we can fit in. Let's just do the questions, and then we just have to have Michelle back to ask more. That's all. I'm always here. That's right. Yeah, we're not going anywhere. I mean, you know, why these people that just like, you didn't get to my question. What? What do you think? You got information we don't? We getting canceled? What do you know? What do you know that we don't know?

You know we can do this again for questions. Yeah, all right. Here we go This is Tristan breaker who says good evening. Dr. Tyson dr. Thaler lord nice Tristan here from Utah Considering gravity is caused by the curvature of spacetime Hypothetically could gravitational waves from two separate sources intercept forming a constructive interference pattern combining their amplitudes if so Could this create regions in space where the curvature of spacetime is great enough to produce a gravitational field as though a celestial body were present even though nothing is there? Additionally, wouldn't the crest of the waves create an inverse gravitational field? Could spacetime curve in the opposite direction, imagining the common fabric of

gravity demonstration, for instance, Producing a repulsive field rather than an attractive one. My heartfelt thanks to you for bringing awareness to the wonders of the universe. Well, Christian, all I could say is, first of all, I need some of that weed. You smoking? Because, man, that is some serious stuff.

Check with our producers. Maybe for questions like that, we should create a little certificate that we send back out to them for questions that are above and beyond the call of duty. We've had two so far in this show. It's a deep thought question, right? That's a deep thought question. Michelle, are you up on gravitational waves? What's the latest thing? Oh, I love gravitational waves. Are you kidding? I was a postdoc at Caltech. I better love gravitational waves. So oh my god, let me just say how impressed I am with gravitational wave detection. Because this is something, when I was a postdoc at Caltech, it was a big deal. The LIGO project, the Laser Interferometric Gravitational Wave Observatory, was something that Caltech was involved in.

Honestly, I talk about my skeptical nature. I was like, they're never gonna detect these bastards. I mean, this is really, so LIGO, at the time, there were two facilities, there are now more. You've got a laser that's about two miles long and another one going out in a 90 degree angle, there we go, in a corner. So two miles each direction. And those lasers are supposed to be exactly the same length. Imagine trying to calibrate that. And so the lasers are going back and forth, bouncing around. And then if a gravitational wave comes by, a gravitational wave is literally a wave in space and time. And so space contracts in one direction, and all of a sudden the two lasers are no longer the same length. They're different because space contracted.

And so the person with the question is absolutely right. That's what gravitational waves do. The thing that makes it mind blowing that we detected these things is they're tiny, luckily, for us. I mean, they're a thousand times smaller than a proton. And to give you a sense of that, I mean, not that the human brain can even get that. And the human brain's not going to get the next one either. But it's the equivalent of measuring from the dust to the nearest star, Alpha Centauri. about four light years, light years about six trillion miles. So you're going on 24 trillion miles. That

would be the equivalent of the distance between us and the nearest star varying by the thickness of a human hair. Wow. They're trying to measure that. Holy. Are you allowed to swear on this book? I'll do it for you. Holy shit.

I could go more than that probably. I could swear I'll do that. You know, I did not think they'd be able to do this. And in fact, Michelle, the Nobel Prize, which was ultimately given to the project, for me, I view that as not only a prize for the scientific result, but for the engineering that enabled it. Oh my gosh. Yeah. With that measurement being so infinitesimally small, How did they know that somebody didn't just bump the table? Absolutely. So, I mean, so this is the thing, right? I mean... Well, breathe on the table. Somebody bumped on table on the next stake, right? It's nearly in Australia. There's all this noise in the detector. Like you said, there's all kinds of noise going on in the detector. Then this, I was just making sure I had my dates and everything. This would have been in September of 2015. Just to clarify, when a scientist uses the word noise, Yeah. Necessarily and almost hardly ever are talking about sound. Yeah, they're not going. Shut your ass up. Who knows? Maybe this is what the detective was saying. Shut down the noise. No, we're talking about sources of interference to the signal you're trying to measure, no matter what those sources are. Right. So go on, Michelle. There's all kinds of little variations. I mean, the lasers are under vacuum, but there's all kinds of things happening.

So in September of 2015, there's all kinds of noise. There's all kinds of variation. But then all of a sudden, one of the detectors goes, wum, bum, bum, bum, bum, bum, boom, just like that. And then at the speed of light between one detector, which was in Louisiana and the other one was in Oregon, the same pattern, wum, bum, bum, bum, bum, bum, bum, boom, goes off. And it was absolutely unquestionable. It was the same pattern, the same frequency at the speed of light. Gravitational waves travel at the speed of light from one detector and people are like Nobel Prize. Yeah, that's fantastic. Oh my god. But let me just say because I want to say what that event was. There were two black holes, 1.4 billion light years away. This event happened 1.4 billion years ago. The black holes were both on the order. In another galaxy. In another galaxy. Yeah, it was way out of our galaxy. Far away by the way.

Far, far away. Thank you, Chuck. Those two black holes are both about... It's far, far away. It's far away. Yeah, yeah. 1.4 billion light years don't need to worry about it. Yeah, that's right. You know, there were two black holes that were roughly 30 times the mass of the Sun up a piece. We caught them in only the last two seconds, 0.2, two tenths of a second, of them spiraling together and merging into a bigger black hole. And in that 0.2 seconds, They accelerated from 30% the speed of light to 60% the speed of light. Black holes 30 times the mass of the sun. And about two times the mass of the sun was converted into pure energy. In .2 seconds. This is an unbelievable thing. It's real. That is a monster of proportions that the human brain can't go any... I have no idea what that is. And it happened and we measured it. And Michelle, wasn't that the...

most energetic event in the universe in that moment, I think it had to be. It was more energy than every star in every galaxy in the observable universe. Wow. Yeah. Yeah. In that .2 seconds, right? In that .2 seconds. Yeah. Jeez Louise. Okay, so tell us about, so if this happens with some frequency, no pun intended there, then two waves that do intersect, what can we say about them? Do they Do they magnify any kind of gravitational disturbance in the space time? What can we say about what is happening? Because when you drop two pebbles in a pond, you see interference patterns with the peak it's higher and the valleys get lower. So it's exaggerates what's there. Presumably, we can expect just that same thing, can't we? Yeah, I think the answer is yes, but it's at a tiny, tiny, tiny scale.

And also this was a fantastic energetic event that the gravitational wave detector was able to detect this event. But there's also a background of gravitational waves, binary stars going around each other, all kinds of things. I'm actually making tiny little gravitational waves right now. So again, we come to this idea of noise. But you were making gravitational waves because you were lifting your hands up and down. A mass is moving. There's something moving. There's mass moving around. That's right. Really, really small. Is that thing you have special powers? You're just moving your body just to be clear. Exactly. Yeah. That'd be a great superpower, you know, making gravitational waves. But anyway, yeah, so the answer I think is yes. But I don't think there's going to be any organized effect of that. Because there's just, I mean, think about, you know, waves on

a giant lake, a giant ocean. There's ripples going every which way. And yes, they are interfering. When they add together, or they can, I think all that's happening. But I don't think that has any measurable effect on what's going on in the universe. It's tiny. So in Tristan's question, it was, would it be able to perhaps give the appearance of a celestial body being there when none is there because of the curvature of space time itself? And you're saying that that's not going to happen.

Okay. Cool. The energetics are too small. So just to be clear, it gave off more energy than all stars combined in the universe over those .2 seconds. But then that energy in the form of gravitational waves moved how many billions of miles to reach us? 1.4 billion light years. 1.4 billion light. So it's been diluted over that distance to the low energies that were ultimately detected in LIGO. I mean just think of the volume. Because anybody at this distance in any direction from those colliding black holes could make this measurement. So that energy is expanding through the entire universe and we're just catching our one little piece of it. What was that event like? If you're close to that event, I mean that must have ripped apart any matter.

you know, within quite a quite a large distance. I mean, I'm sure we could, you know, have somebody do the math on that. So I mean, it's, it's, it's, no, I don't think that there would be, it can't make it seem like there's a star there. This is not a solution for dark matter. Sorry. All right. This is Rachel Ambrose who says Rachel here from Texas. I loved Michelle's Big Think interview. It was the first time I heard someone articulate the profound weirdness of reality in a way that completely resonated with how I have always felt, but could never adequately explain it. I just appreciated that so much. Thank you, Michelle. My question is about neutron stars. I am confused why the high-energy fast radio bursts the FRBs.

come to us in the form of low energy radio frequencies rather than x-rays or gamma rays. What mechanism causes such a high energy event to manifest as radio light? So this has built in the knowledge that radio waves are lower energy light than x-rays or gamma rays. And if it's a high energy phenomenon, why is it wasting itself with radio waves? Okay, so yeah, I at least know some a little bit about that I mean as I said I mean one of the missions I absolutely adore is called nicer the neutron star interior

composition Explorer. It's on the space station. It's a Goddard run mission Fantastic, it's about the size of a washing machine again not not an expensive mission And it's able to actually get x-rays, in this case, x-rays from neutron stars. And at times the arrival of the x-rays so precisely, we can make a map of what the surface of a neutron star is like. They're tiny. Neutron stars are only about 20 miles across. They usually have about twice the mass of the sun packed into 20 miles. These are extreme things, and they're real. I mean, we study many of these.

And they're so incredible when you do a map of these, and they often are rotating very fast, you actually get this strange effect on the side of the map because light is bending around. There's so much mass, the curvature of space and time, is so extreme. Two solar masses packed into 20 miles that you actually can see behind a neutron star as well as in front, the light curves around it. And so, I mean, what an incredible real monster. So what you're saying is it has many properties that we would find with black holes in its distortion of the fabric of space and time. In fact, the neutron star group at Goddard said that neutron stars are far more interesting than black holes. Because with black holes, you have this dark event horizon that nothing ever comes out of. You don't know what's going on in there. With neutron stars, you got the fricking thing right in front of you. It's very bright. I mean, they still are here with us. And you can study something.

that has such extreme conditions that there's no laboratory on earth. We can even study what matters like under these conditions. It's amazing. But yeah, okay, so going back to why it's radio waves. Neutron stars definitely do admit high-energy radiation. They're actually the source of what we think are gamma ray bursts, dangerous blasts of high-energy radiation. We're using an X-ray telescope to observe these things. The surface of a neutron star is very hot. It definitely does generate these high-energy things. The way you get radio waves is something, Neil, I'm sure it's a favorite of yours too. It's a lovely thing called synchrotron radiation. Would you like to go ahead and talk about synchrotron radiation? How could you not? No, as I understood it, the neutron star has to be magnetic, has to have a magnetic field. That's right. And as it rotates, it can accelerate electrons that are ambient in the environment. And you can accelerate electrons off of a magnetic field that will radiate radio waves.

Yes, we call it synchrotron radiation. I think I accounted for that correctly. Is that correct? Yeah, I mean the one thing that's perhaps something to mention about synchrotron radiation. I mean, so you've got a couple things going on. I mean, you also have, you know, electrons are charged particles. They respond to a magnetic field. They can actually be accelerated. The neutron stars have jets. We see those as well. Also, when you get Synchrotron radiation happens largely when electron spirals around a magnetic field. Just by the act of spinning around, that's a form of exploration. That's what it was. It creates its own frequency, spiral frequency that is the radio wave. So what's fun about the neutron star, just to summarize that, is different processes within the environment of a neutron star give you different bands of light that our different telescopes can participate in.

Which is kind of fun and then you can compare notes across different ways. This is why astronomers just aren't greedy. It's like, oh, we need a radio telescope and an infrared telescope and an x-ray telescope and a visible light telescope. But you're absolutely true. You see totally different things in these different wavelengths of light. And so as high energy and dramatic objects neutron stars are, you better believe they are. There was one in God, was it 2007? There was an event that actually blew off a measurable amount of our atmosphere. And the whole magnetic field of the Earth was ringing like a bell for a while. Now you're telling us this? Well, look this up. This is all real. And we traced it back to a neutron star that we think is about 50,000 light years away. The neutron star had a little bit of a bad day. The crest we think shifted by about a centimeter.

And that gave off so much energy that 50,000 light years away, it blew off a bunch of our atmosphere. Oh my. These are intense things. Michelle, I'm going to be honest. I could have lived without that information. Yeah, I know. It keeps you up at night, doesn't it? I'm curious about asteroids, right? I mean, killer asteroids? No. I mean, these blasts of high energy radiation from neutron stars, some people think that may limit the age of civilizations in the galaxy. Eventually, we all could hit that one. Well, another great question. Thank you, Rachel. All right. This is Brett. And Brett says, hi, Dr. Tyson, Dr. Thaler, Lord Nice. Tuning in from Dublin, Ireland. My question

is what unlikely possibility or possibilities do you hope are true? Oh, that's unlikely. That's father for good thought.

It is. An unlikely possibility that you're like, oh man, that'd be so great if that were true. Yeah. So, boy, that can go in so many different directions, because I've got, I mean, one of the things I- What a great question, Brad. Look at you. I can lead all. I got one for you, Michelle. Okay, I got one too. You go ahead. I think I got it. I think it's unlikely that we'll find anything other than microbial life on Mars if we find life at all. but we would be really cool if we found sort of macroscopic crawly things living in the subsurface of Mars. I think that's unlikely, but that would be really cool if creatures crawl out from hole. Would you still be satisfied if those macroscopic crawly things were fossilized in some way, but they're no longer there? No, we're talking about unlikely things that we want to happen. Let the thing crawl out and ride one of the rovers like a...

Like, uh, like, uh, you know, I got. Right. You know, like, like what do you call those horses? Those, the writers in the bar. Yeah. I want to come out and ride one of our ride one of our rovers. Unlikely, but it'd be really cool if it was true. So now your turn, Michelle. Well, I mean, I mean, since I was just riffing on that, I mean, do you think it might be more likely to find macroscopic life on the moons of Jupiter and Saturn in the oceans? you know, the Europa and Salatus. We're talking about unlikely things that we want to happen. Oh, you think that's a likely thing? Okay, all right, got it. Yeah, yeah. Unlikely things. Yeah, there could be like little brine shrimp and stuff. Yeah. One of my big excitements right now with the web telescope is it's looking so far away that we're looking back to a time roughly, like let's say about 300 million years after the Big Bang. And we're actually seeing the very first stars.

form. And they're absolutely different. I mean, we haven't actually found one of these first stars yet. I really want to find a first generation star. We're going to be looking back to a time, just like I said, a couple hundred million years after the Big Bang when stars started to form. Webb can now see that far back. We have that power. And we're starting to see these pseudo stars that are formed around black holes. They're about a million times the mass of the sun, relatively small. I mean, these things may become the seeds of

the big black holes that are in the centers of all galaxies. We're starting to see that forming now. And my question, I was a stellar astrophysicist. I was a stellar astrophysicist. I studied stars. That was my thing. Outstanding in your field. Not outstanding, but that's why I studied. And I would love to see what one of the first generation stars looks like, because my guess is it's going to look totally different.

The universe was more dense back then. There was more dark matter prevalent. Dark matter has gravity. That's going to get sucked into the formation of a star just for the gravity, too. What was that first generation of stars like? My guess is they're going to be huge, many, many times the mass of the sun and many, many times bigger than the most massive stars we have today. And I'm wondering if they're going to work really differently because there's more dark matter in them. I want to see that first generation of stars. And just to be clear, James Webb was specifically designed and tuned for that exercise to see the beginnings of the formation of matter in the early universe. So you're betting on the right horse there, I think.

Yeah, we're doing it. I mean, never let astronomers name anything. These things are called little red dots. Ha-ha, yeah. Well, they're massive pseudo stars around a giant black hole at the beginning of the universe, what we're calling little red dots. Because they look like little red dots. Little red dots, yeah. But they're so far away. That's one of my favorites. Chuck, time for like two more questions. All right, here we go. But only if Michelle is efficient in her answers. I promise to be efficient. You're on notice. You don't have to promise anything, Michelle. It's going to do what you want. This is Alyssa Feldhaus who says, hello, Fowler the Great. This is Alyssa Feldhaus tuning in from Hendersonville, Tennessee as such a prominent female figure in the science world. What message can you give?

all the little girls out there that will one day stand on our shoulders. Me and my girls love how the universe works, and thank you for keeping science cool. I always think of myself as so nerdy and I never think about cool. Okay, so a message I always have. The nerdy is the new cool. Nerdy is the new cool. Nerdy is the new cool, absolutely. A message I already have is you're already the right type of person to be a scientist. Math and physics did not come quickly to me. I didn't get particularly good grades in school

in math and physics. It took me some time. I just loved it so much. Paying attention to your own curiosity and not being discouraged when maybe you don't get something very, very quickly. You're already enough, you're already there. For so long I wondered, do I have what it takes to be an astrophysicist? I love this, but maybe I just don't have the magic thing, the right stuff. And science doesn't work that way.

If you want to study it, you can study it. It's like learning a language. It takes a while to become fluent in a different language. You don't really go up to a young person and say, you could never learn Spanish, even if you tried. No, no. I mean, science is a beautiful thing to study. It takes time. And anyone can learn it if they're interested in it and have that passion more than any innate ability and talent to get all of this very quickly. So, you know, I spent way too much time thinking maybe I'm not the right personality. I'm not smart enough. And it's a cultural, you know, bullshit that science requires a certain type of brilliant, different person. So, you know, you're enough and, you know, you're going to be the best scientist.

From your perspective, right? I mean we're all people have different science talents. Some people are better at the math. Some people see the larger concepts. Some people are really good at communicating it. These are all really important steps and you can find that part that you fit into. So you don't have to be anything other than just the way you are. Wow, look at that. And if I can add... not having ever been female, but being a black man in the world. There are resonant challenges that we face entering a world that role models are not there. So I would just add that there are always people who will learn something faster than you will. And the school system tends to reward that as and call those people smart and they might get a higher grade. And but in the end, That's not what matters. What matters is your motivation to ultimately learn what you need to learn. If it takes you longer to learn it and longer than the time before you test it on it, it takes you a little longer. So what took you a little longer? If you get there and your ambition takes you and your interest, then ultimately that's what will dominate who and what you are as a participant on the research frontier. Not how quickly you learn a problem set on an exam.

in high school or in college. So ambition is something that tests hardly ever test for. And that's what you need. And that's what you need confidence in. Yeah. And just being able to stick to something. Yeah. And just having the grit. I mean, when I got to college, I got placed in remedial math. I came from a public school system that didn't have a whole lot of math. And, you know, I just knew I could not get space out of my head. I was so fascinated by it and I am so glad I stuck in when I was thinking, geez, I just don't have the talent for this. And that became the driver, your source of energy to pursue it. And I like your language analogy because no one will ever say, oh, you're not smart enough to learn Spanish. No one will ever say that. They just say, are you committed enough to learn Spanish? And you learn a little bit today and a little bit next week and next month. And eventually you wake up and you're just talking fluent Spanish.

you know, and everyone can learn it. And I think that's also true with math as a language of the universe. Yes. Well, let me add, Alyssa, that as a product of Philadelphia public schools, there's always colony. All right. So, Chuck, we only actually had time for that one last question, but that was a really good one to end this episode on. So, Michelle, thank you for being a guest on StarTalk. Hey, I'm always here. Where are you based right now? Right now in Milwaukee, Wisconsin. Whoa, is that where you're based? Yes, yes. Cool, cool. Okay. Well, we will make sure to find you again because you're a delight. to share in our answers here and to get your perspectives on the state of science in America and in the universe. All right. Chuck, always good to have you, man. Always a pleasure. These are fan favorites, these cosmic queries. And we delight in the questions that we receive and keep them coming. But you have to be a member of the Patreon community first. Yes, but it's only five dollars.

That's it. 5 a month. And I gotta tell you, that's less than what the budget of NASA is. All right. I am Neil deGrasse Tyson, your personal astrophysicist. As always, I bid you to keep looking up.